Patient Readmission Risk Analysis Report

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1. Project Overview

This report presents an end-to-end analysis of diabetic patient readmission risk using real hospital data from 130 US hospitals. The goal is to predict whether a patient will be readmitted within 30 days of discharge using machine learning.

2. Dataset Summary

Metric	Value
Total Patients	101,766
Total Features	43
Readmitted Patients	11,357
Not Readmitted	90,409
Readmission Rate	11.16%
Average Hospital Stay	4.40 days
Average Medications	16.02
Average Lab Procedures	43.10
Average Diagnoses	7.42

3. Key Findings

• Patients with more inpatient visits are most likely to be readmitted

• Readmitted patients take significantly more medications (p-value < 0.05)

• Average hospital stay is 4.40 days across all patients

• Female patients slightly outnumber male patients in this dataset

• Only 11.16% of patients were readmitted within 30 days

• Number of diagnoses strongly correlates with number of medications

• Elderly patients (60-70) show highest readmission rates

• Patients with more emergency visits have higher readmission risk

4. Statistical Analysis

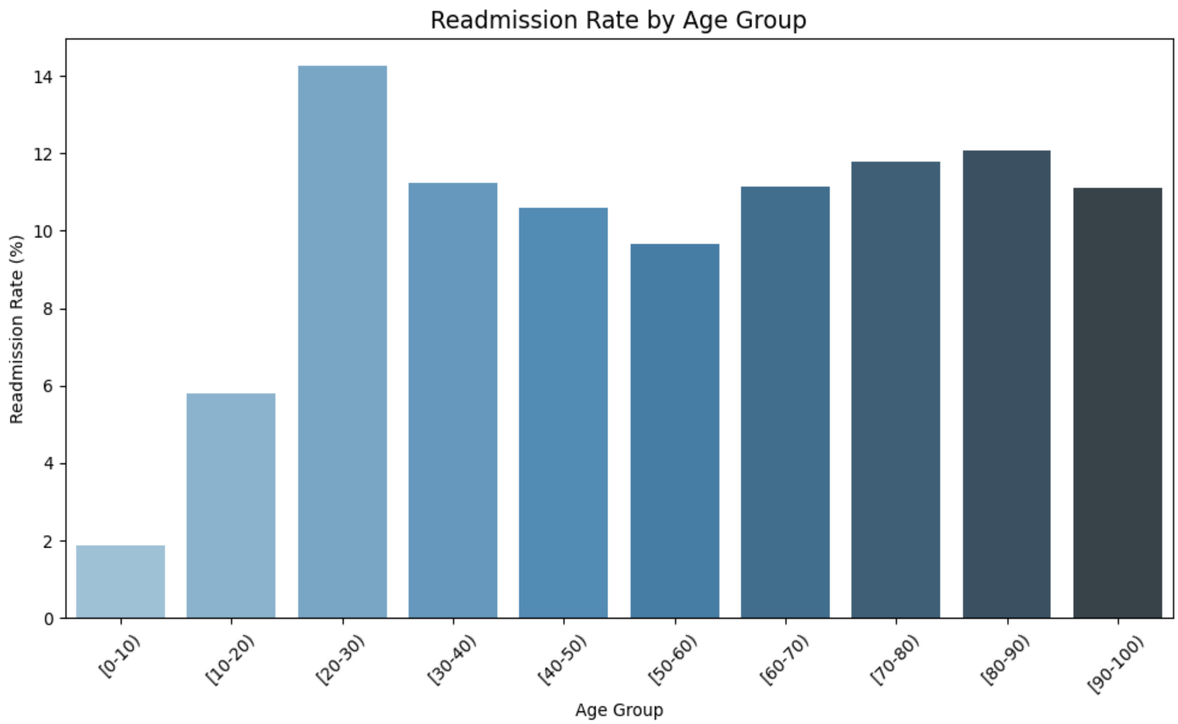
Metric	Value
Hypothesis Test	Independent T-Test
T-Statistic	12.2690
P-Value	0.0000
Result	Significant difference found
Time in Hospital Skewness	1.1340
Medications Skewness	1.3267
Lab Procedures Skewness	-0.2365

5. Model Performance

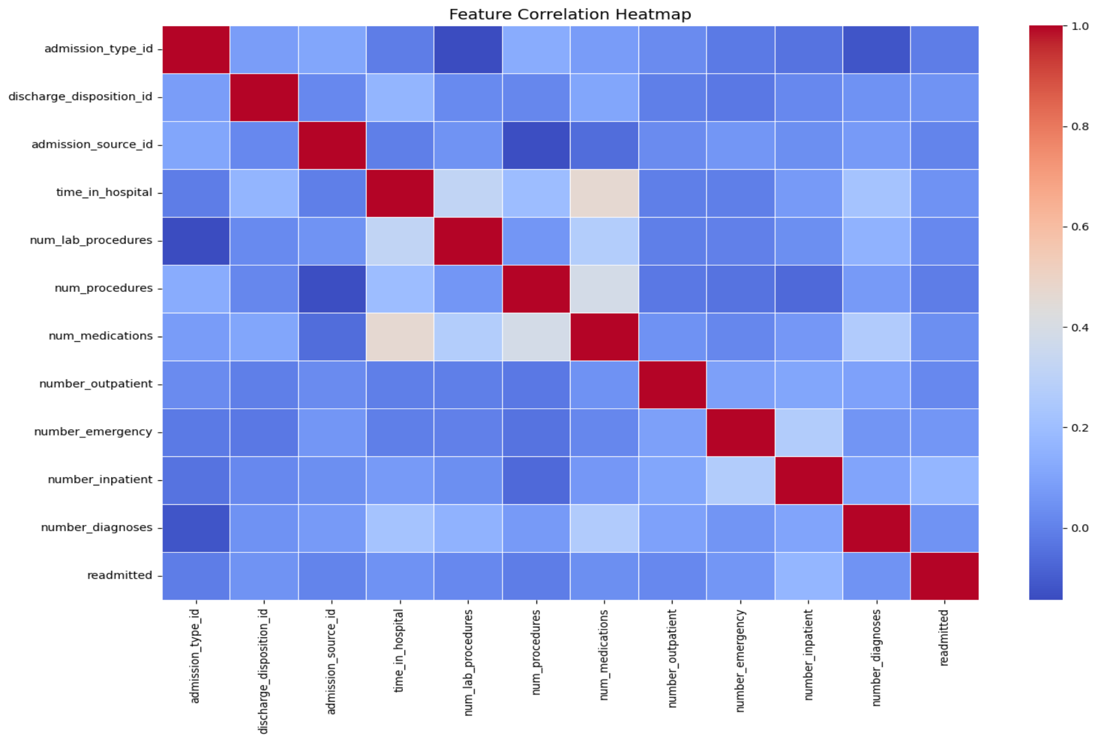
Metric	Value
Model Used	Logistic Regression
Training Patients	81,412
Testing Patients	20,354
Overall Accuracy	66.89%
Recall (Readmitted)	52%
ROC-AUC Score	0.6464

6. Visualizations

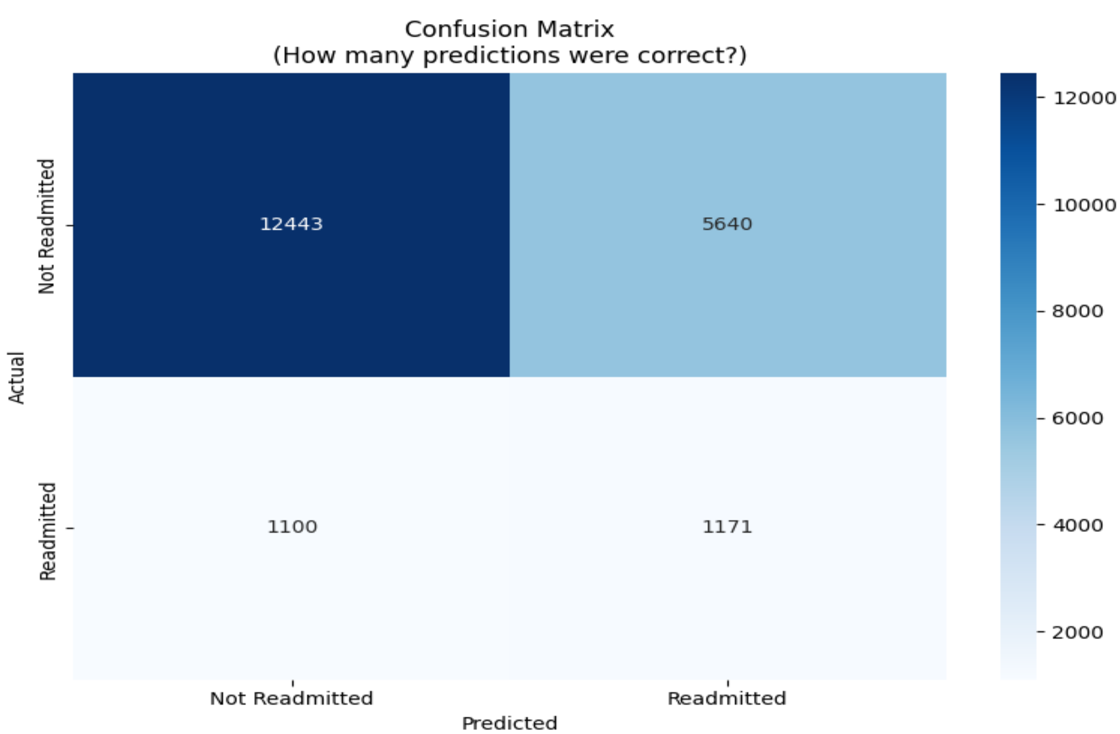
Readmission Rate by Age Group



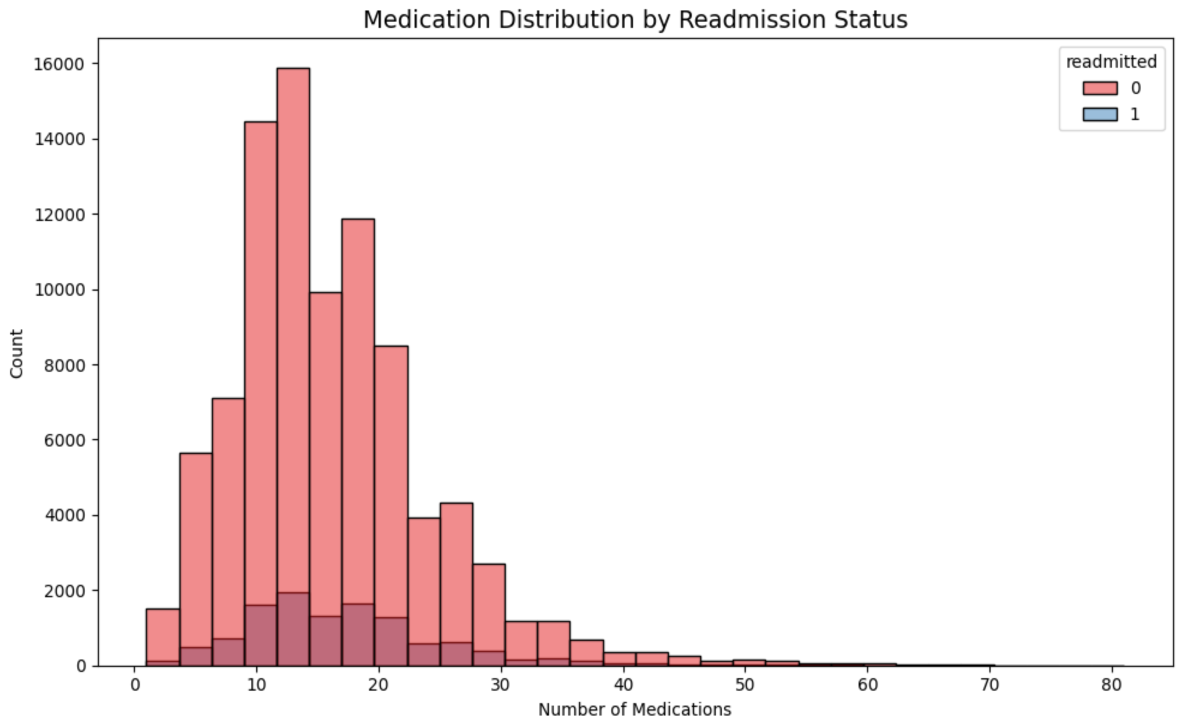
Feature Correlation Heatmap



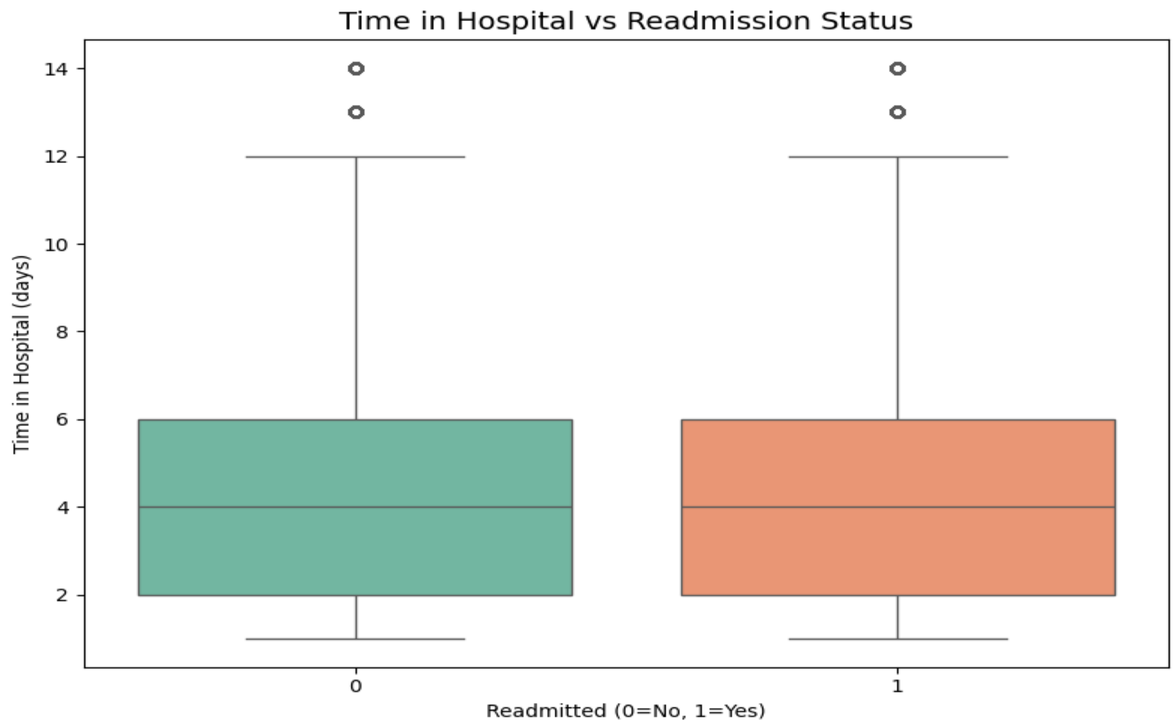
Model Confusion Matrix



Medication Distribution



Time in Hospital vs Readmission


Time in Hospital vs Readmission Status
Time in Hospital (days)
14
12
10
8
6
4
2
0
1
Readmitted (0=No, 1=Yes)